

Group 6 — Project Submission Report

A Comparative Framework for Survival Analysis on the RMS Titanic
Naïve Bayesian vs. Decision Tree Classifiers

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DSC 523 — Data Mining | Instructor: Dr. Anu Maria Sebastian

25 April 2026

1. Project Repository

The complete project — including the Jupyter notebook, all source scripts, the Kaggle `train.csv` dataset, every figure used in the conference paper, and the compiled paper itself — is publicly hosted at:

<https://github.com/vtsarin/dsc-523-mini-project>

The conference paper (PDF) and the executable notebook are reproducible end-to-end from the scripts `run_analysis.py`, `build_notebook.py`, and `make_flowchart.py`; all stochastic components use `random_state = 42`.

2. Member Contribution Table

Table 1: Group 6 — Member contributions.

Member's Name		Sections Written	Tasks Undertaken
Sarin Thayyil	Vadakke	§1 Introduction; §4 Methodology; §6 Implications and Applications; §7 Conclusion (final consolidation)	Project management and team coordination; defined the 80/20 stratified evaluation protocol; designed the comparative experimental framework; authored and assembled the final manuscript; managed deliverables and GitHub repository.
Lekshmi S S		§2 Literature Review (all five subsections including Research Gap and Motivation)	Comparative analysis of Gini vs. entropy impurity measures; literature survey of NB, DT, LR, RF on the Titanic benchmark; identified the research gap; framed the theoretical foundations.
Akbar T E		Figure captions and visual narrative across §3, §4, §5	Produced all 11 figures (KDD flowchart, EDA panels, age-fare distribution, correlation heatmap, confusion matrices, ROC curves, feature importances, decision-tree visualisation); designed the consistent colour palette and matplotlib styling; built <code>make_flowchart.py</code> .
Suja V		§3 Dataset Description; §4.1 Data Profiling; §4.2 Missing-Value Treatment; §4.3 Feature Engineering; §4.4 Encoding	Designed the group-wise (<code>Pclass</code> × <code>Sex</code>) median imputation rule for <code>Age</code> ; mined the <code>Cabin</code> field for the <code>Deck</code> feature; engineered <code>Title</code> , <code>FamilySize</code> , <code>IsAlone</code> , <code>AgeBin</code> and <code>FareBin</code> ; built the encoding pipeline and verified zero residual missingness.
Nikhil Ravi		§5 Results and Discussion (including §5.2 Model Performance, §5.3 Statistical Significance, §5.4 Discussion)	Implemented and ran all four classifiers; computed 10-fold stratified CV scores; verified every confusion matrix; ran the McNemar exact paired test ($p = 8.22 \times 10^{-6}$); produced <code>results.json</code> ; reviewed manuscript for compliance with course guidelines.

3. Visualisations

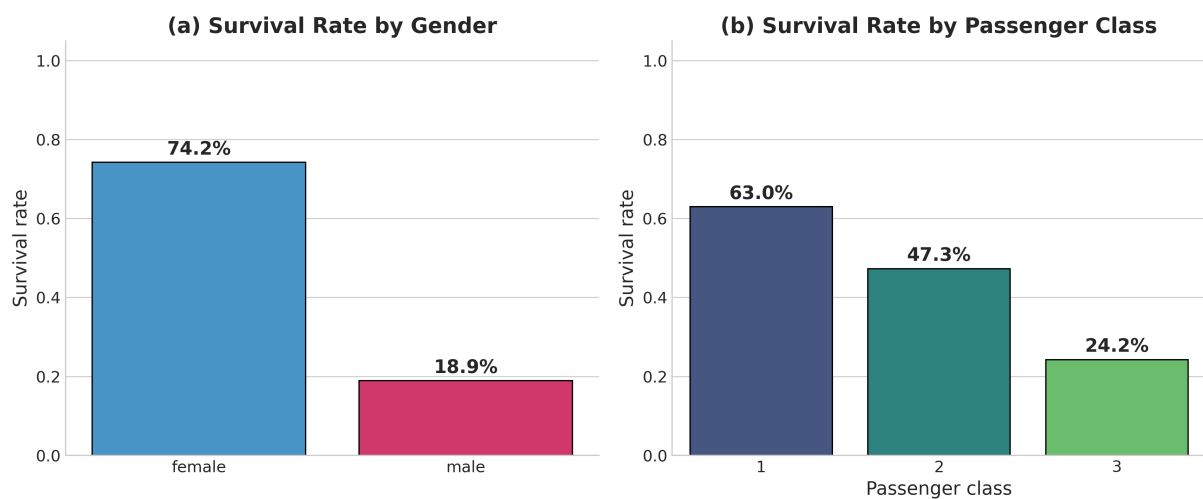


Figure 1: Univariate survival rates by the two strongest predictors. Female passengers survived at 74.2% versus 18.9% for males; survival declines monotonically with passenger class.



Figure 2: Joint effect of class and gender. First-class women survived at 96.8%; third-class men survived at only 13.5%.

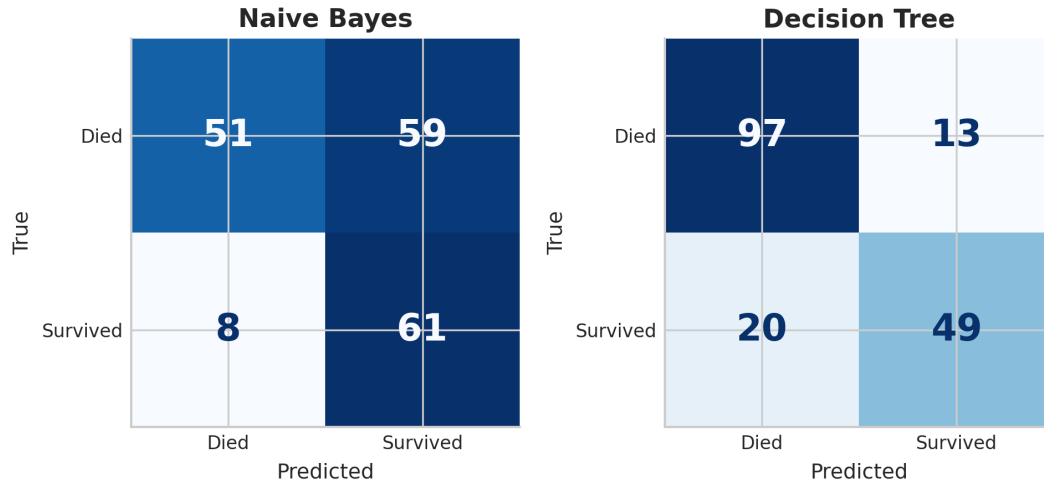


Figure 3: Confusion matrices on the held-out test set. Naïve Bayes over-predicts the positive class (59 false positives); the Decision Tree is materially better balanced (only 13 false positives).

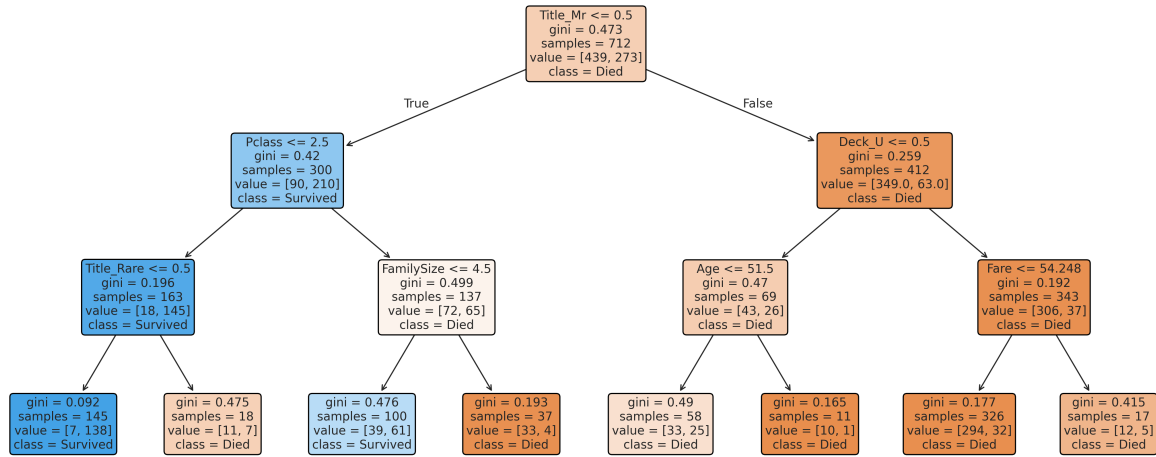


Figure 4: Fitted Decision Tree (`max_depth=3` shown for readability; deployed model uses depth 4). The root split on `Title_Mr` captures the bulk of the gender / honorific signal in a single question.

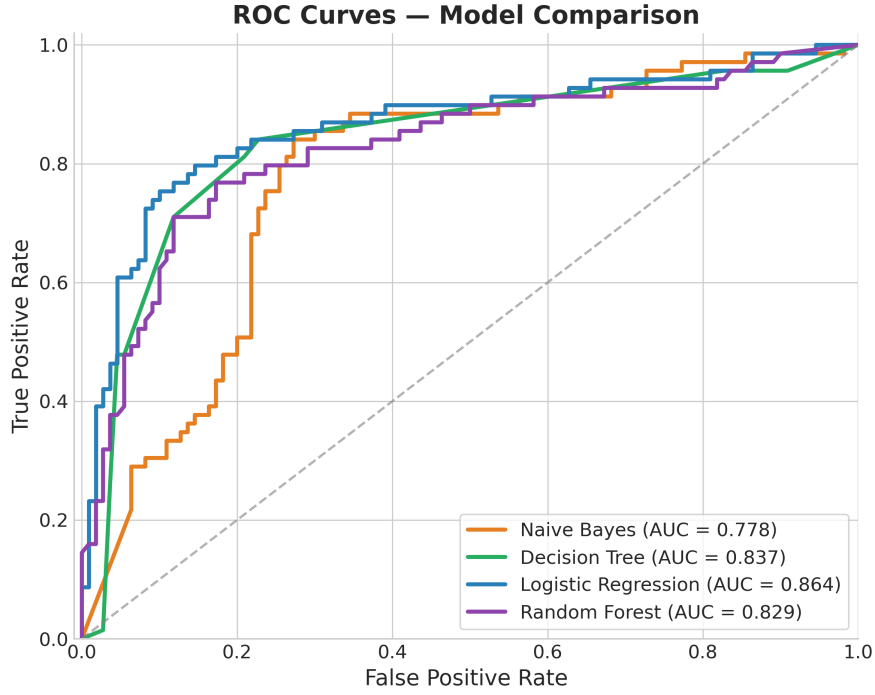


Figure 5: ROC curves for all four classifiers. Logistic Regression dominates slightly across most thresholds (AUC = 0.864); Naïve Bayes trails uniformly.

4. Most Important Findings

1. **The Decision Tree decisively outperforms Naïve Bayes by ~19 percentage points.** On the held-out test set ($n_{\text{test}} = 179$), the Decision Tree achieves 81.56% accuracy (AUC = 0.838) while Gaussian Naïve Bayes reaches only 62.57% (AUC = 0.778). McNemar’s exact paired test gives $p = 8.22 \times 10^{-6}$, rejecting the null of equal error rates at any conventional level — the gap is not a sampling artefact.
2. **Logistic Regression is the strongest single model overall.** After feature engineering, the decision boundary is close to linear: Logistic Regression attains 83.24% test accuracy and AUC = 0.864. Random Forest (81.01%) does not outperform the single Decision Tree, suggesting the engineered feature set already captures the important interactions.
3. **Gender dominates; class modulates.** Female passengers survived at 74.2% versus 18.9% for males ($\chi^2 = 260.7$, $p = 1.2 \times 10^{-58}$). Survival declines monotonically with class: 1st (63.0%), 2nd (47.3%), 3rd (24.2%). The interaction is striking — 1st-class women survived at 96.8% (91 of 94) while 3rd-class men survived at only 13.5% (47 of 347).
4. **Family size has a non-monotonic effect.** Survival peaks at family sizes of 2–4 (55–72%) and collapses for solo passengers (30.4%) and very large families of ≥ 5 (16%) — a pattern neither `SibSp` nor `Parch` captures alone, justifying the engineered `FamilySize` feature.
5. **NB fails because multicollinearity violates its independence assumption.** After feature engineering, several feature pairs are strongly correlated by construction: $r(\text{SibSp}, \text{FamilySize}) = +0.89$, $r(\text{Sex_female}, \text{Title_Mr}) = -0.87$, $r(\text{Parch}, \text{FamilySize}) = +0.78$. These violate the class-conditional independence assumption directly, compounding the per-feature evidence terms and distorting the NB posterior — the textbook failure mode of Gaussian NB on engineered tabular data.